MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYTETRAFLUOROETHYLENE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -67°C (-88°F) to +250°C (+482°F).

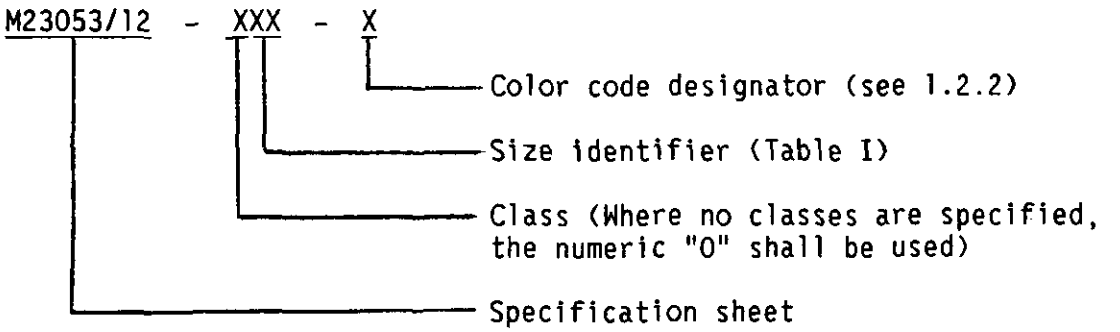
Class: The sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Thick wall sleeving
Class 2 - Standard wall sleeving
Class 3 - Thin wall sleeving
Class 4 - Very thin wall sleeving

Color: The sleeving shall be furnished in clear (transparent milk white to light tan) or in the color specified by the acquiring activity. Colors shall conform to the requirements of Class I of MIL-STD-104.

Longitudinal change: ± 20 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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Example: Thin walled (Class 3), 0.430 inch (10.9 mm) as supplied ID, clear sleeving shall be identified as M23053/12-327-C.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/12-101-*	.166(4.22)	.130(3.30)	.030 ± .005(.762 ± .127)
M23053/12-102-*	.250(6.35)	.193(4.90)	.030 ± .005(.762 ± .127)
M23053/12-103-*	.333(8.46)	.257(6.53)	.030 ± .005(.762 ± .127)
M23053/12-104-*	.415(10.54)	.320(8.13)	.030 ± .005(.762 ± .127)
M23053/12-105-*	.498(12.65)	.383(9.73)	.030 ± .005(.762 ± .127)
M23053/12-106-*	.580(14.73)	.448(11.38)	.030 ± .006(.762 ± .152)
M23053/12-107-*	.666(16.92)	.510(12.95)	.030 ± .006(.762 ± .152)
M23053/12-108-*	.748(19.00)	.572(14.53)	.030 ± .006(.762 ± .152)
M23053/12-109-*	.830(21.08)	.637(16.18)	.030 ± .006(.762 ± .152)
M23053/12-110-*	.915(23.24)	.700(17.78)	.032 ± .006(.813 ± .152)
M23053/12-111-*	1.000(25.40)	.764(19.41)	.040 ± .007(1.02 ± .178)
M23053/12-112-*	1.170(29.72)	.891(22.63)	.045 ± .007(1.14 ± .178)
M23053/12-113-*	1.330(33.78)	1.020(25.91)	.050 ± .008(1.27 ± .203)
Class 2			
M23053/12-201-*	.050(1.27)	.027(.069)	.012 ± .002(.305 ± .051)
M23053/12-202-*	.055(1.40)	.032(0.81)	.012 ± .002(.305 ± .051)
M23053/12-203-*	.060(1.52)	.039(0.99)	.016 ± .003(.406 ± .076)
M23053/12-204-*	.065(1.65)	.043(1.09)	.016 ± .003(.406 ± .076)
M23053/12-205-*	.076(1.93)	.049(1.24)	.016 ± .003(.406 ± .076)
M23053/12-206-*	.085(2.16)	.054(1.37)	.016 ± .003(.406 ± .076)
M23053/12-207-*	.110(2.79)	.067(1.70)	.016 ± .003(.406 ± .076)
M23053/12-208-*	.120(3.05)	.072(1.83)	.016 ± .003(.406 ± .076)
M23053/12-209-*	.130(3.30)	.080(2.03)	.016 ± .003(.406 ± .076)
M23053/12-210-*	.140(3.56)	.080(2.03	.016 ± .003(.406 ± .076)
M23053/12-211-*	.150(3.81)	.089(2.26)	.016 ± .003(.406 ± .076)
M23053/12-212-*	.170(4.32)	.101(2.57)	.016 ± .003(.406 ± .076)
M23053/12-213-*	.191(4.85)	.112(2.84)	.016 ± .003(.406 ± .076)
M23053/12-214-*	.205(5.21)	.124(3.15)	.020 ± .004(.508 ± .102)
M23053/12-215-*	.215(5.46)	.130(3.30)	.020 ± .004(.508 ± .102)
M23053/12-216-*	.240(6.10)	.141(3.58)	.020 ± .004(.508 ± .102)
M23053/12-217-*	.270(6.86)	.158(4.01)	.020 ± .004(.508 ± .102)
M23053/12-218-*	.302(7.67)	.178(4.52)	.020 ± .004(.508 ± .102)
M23053/12-219-*	.320(8.13)	.198(5.03)	.020 ± .004(.508 ± .102)
M23053/12-220-*	.370(9.40)	.224(5.69)	.020 ± .004(.508 ± .102)
M23053/12-221-*	.390(9.91)	.249(6.32)	.020 ± .004(.508 ± .102)
M23053/12-222-*	.410(10.41)	.260(6.60)	.020 ± .004(.508 ± .102)
M23053/12-223-*	.430(10.92)	.278(7.06)	.020 ± .004(.508 ± .102)
M23053/12-224-*	.450(11.43)	.311(7.90)	.020 ± .004(.508 ± .102)
M23053/12-225-*	.470(11.94)	.329(8.36)	.020 ± .004(.508 ± .102)

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/12-226-*	.470(11.94)	.347(8.81)	.020 ± .004(.508 ± .102)
M23053/12-227-*	.470(11.94)	.334(8.48)	.025 ± .006(.635 ± .152)
M23053/12-228-*	.560(14.22)	.399(10.13)	.025 ± .006(.635 ± .152)
M23053/12-229-*	.655(16.64)	.462(11.73)	.025 ± .006(.635 ± .152)
M23053/12-230-*	.750(19.05)	.524(13.31)	.025 ± .006(.635 ± .152)
M23053/12-231-*	.930(23.62)	.655(16.64)	.030 ± .006(.762 ± .152)
M23053/12-232-*	1.125(28.58)	.786(19.96)	.035 ± .008(.889 ± .203)
M23053/12-233-*	1.310(33.27)	.911(23.14)	.035 ± .008(.889 ± .203)
M23053/12-234-*	1.500(38.10)	1.036(26.31)	.035 ± .008(.899 ± .203)
Class 3			
M23053/12-301-*	.034(0.86)	.015(0.38)	.009 ± .002(.229 ± .051)
M23053/12-302-*	.038(0.97)	.018(0.46)	.009 ± .002(.229 ± .051)
M23053/12-303-*	.046(1.17)	.022(0.56)	.010 ± .003(.254 ± .076)
M23053/12-304-*	.050(1.27)	.027(0.69)	.010 ± .002(.254 ± .076)
M23053/12-305-*	.055(1.40)	.032(0.81)	.012 ± .003(.305 ± .076)
M23053/12-306-*	.060(1.52)	.039(0.99)	.012 ± .003(.305 ± .076)
M23053/12-307-*	.065(1.65)	.043(1.09)	.012 ± .003(.305 ± .076)
M23053/12-308-*	.076(1.93)	.049(1.24)	.012 ± .003(.305 ± .076)
M23053/12-309-*	.085(2.16)	.054(1.37)	.012 ± .003(.305 ± .076)
M23053/12-310-*	.093(2.36)	.061(1.55)	.012 ± .003(.305 ± .076)
M23053/12-311-*	.110(2.79)	.067(1.70)	.012 ± .003(.305 ± .076)
M23053/12-312-*	.120(3.05)	.072(1.83)	.012 ± .003(.305 ± .076)
M23053/12-313-*	.140(3.56)	.080(2.03)	.012 ± .003(.305 ± .076)
M23053/12-314-*	.150(3.81)	.089(2.26)	.012 ± .003(.305 ± .076)
M23053/12-315-*	.153(3.89)	.096(2.44)	.012 ± .003(.305 ± .076)
M23053/12-316-*	.170(4.32)	.101(2.57)	.012 ± .003(.305 ± .076)
M23053/12-317-*	.191(4.85)	.112(2.84)	.012 ± .003(.305 ± .076)
M23053/12-318-*	.205(5.21)	.124(3.15)	.015 ± .004(.381 ± .102)
M23053/12-319-*	.215(5.46)	.130(3.30)	.015 ± .003(.381 ± .076)
M23053/12-320-*	.240(6.10)	.141(3.58)	.015 ± .004(.381 ± .102)
M23053/12-321-*	.270(6.86)	.158(4.01)	.015 ± .004(.381 ± .102)
M23053/12-322-*	.302(7.67)	.178(4.52)	.015 ± .004(.381 ± .102)
M23053/12-323-*	.320(8.13)	.198(5.03)	.015 ± .004(.381 ± .102)
M23053/12-324-*	.370(9.40)	.224(5.69)	.015 ± .004(.381 ± .102)
M23053/12-325-*	.390(9.91)	.249(6.32)	.015 ± .004(.381 ± .102)
M23053/12-326-*	.410(10.41)	.260(6.60)	.015 ± .004(.381 ± .102)
M23053/12-327-*	.430(10.92)	.278(7.06)	.015 ± .004(.381 ± .102)
M23053/12-328-*	.450(11.43)	.311(7.90)	.015 ± .004(.381 ± .102)
M23053/12-329-*	.470(11.94)	.329(8.36)	.015 ± .004(.381 ± .102)
M23053/12-330-*	.470(11.94)	.347(8.81)	.015 ± .004(.381 ± .102)

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 4			
M23053/12-401-*	.050(1.27)	.025(0.64)	.006 ± .002(.152 ± .051)
M23053/12-402-*	.055(1.40)	.031(0.79)	.006 ± .002(.152 ± .051)
M23053/12-403-*	.060(1.52)	.038(0.97)	.006 ± .002(.152 ± .051)
M23053/12-404-*	.065(1.65)	.043(1.09)	.006 ± .002(.152 ± .051)
M23053/12-405-*	.076(1.93)	.046(1.17)	.006 ± .002(.152 ± .051)
M23053/12-406-*	.085(2.16)	.054(1.37)	.006 ± .002(.152 ± .051)
M23053/12-407-*	.093(2.36)	.057(1.45)	.006 ± .002(.152 ± .051)
M23053/12-408-*	.110(2.79)	.063(1.60)	.006 ± .002(.152 ± .051)
M23053/12-409-*	.120(3.05)	.072(1.83)	.008 ± .002(.203 ± .051)
M23053/12-410-*	.140(3.56)	.080(2.03)	.008 ± .002(.203 ± .051)
M23053/12-411-*	.150(3.81)	.089(2.26)	.008 ± .002(.203 ± .051)
M23053/12-412-*	.170(4.32)	.099(2.51)	.008 ± .002(.203 ± .051)
M23053/12-413-*	.191(4.85)	.110(2.79)	.008 ± .002(.203 ± .051)
M23053/12-414-*	.205(5.21)	.122(3.10)	.008 ± .002(.203 ± .051)
M23053/12-415-*	.215(5.46)	.130(3.30)	.008 ± .002(.203 ± .051)
M23053/12-416-*	.240(6.10)	.139(3.53)	.008 ± .002(.203 ± .051)
M23053/12-417-*	.270(6.86)	.154(3.91)	.008 ± .002(.203 ± .051)
M23053/12-418-*	.302(7.67)	.172(4.37)	.010 ± .003(.254 ± .076)
M23053/12-419-*	.320(8.13)	.192(4.88)	.010 ± .003(.254 ± .076)
M23053/12-420-*	.370(9.40)	.214(5.44)	.010 ± .003(.254 ± .076)
M23053/12-421-*	.390(9.91)	.241(6.12)	.010 ± .003(.254 ± .076)
M23053/12-422-*	.410(10.41)	.260(6.60)	.010 ± .003(.254 ± .076)
M23053/12-423-*	.430(10.92)	.270(6.86)	.010 ± .003(.254 ± .076)
M23053/12-424-*	.450(11.43)	.301(7.65)	.010 ± .003(.254 ± .076)
M23053/12-425-*	.470(11.94)	.329(8.36)	.012 ± .003(.305 ± .076)
M23053/12-426-*	.470(11.94)	.347(8.81)	.012 ± .003(.305 ± .076)

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The asterisk (*) in the part number shall be replaced by color code designators (see 1.2.2).

Unrestricted shrinkage: Test method 4.6.5; 350° ± 4°C (662° ± 8°F) for 10 minutes.

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TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 400° ± 5°C (752° ± 9°F)
Restricted shrinkage	Pass	4.6.6.1.2 350° ± 4°C (662° ± 7.2°F), 5 minutes
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	2,500(17.3)	4.6.14 ASTM D 638, *2 inches/minute*
Ultimate elongation, percent, minimum	200	4.6.14 ASTM D 638, 2 inches/minute
Dielectric strength, volts/mil (Kv/mm), minimum	800(31.5)	4.6.2 ASTM D 2671
Volume resistivity, Ohms-cm, minimum	10^{18}	4.6.2 ASTM D 876
Water absorption, percent, maximum	0.01	4.6.2 ASTM D 570, para. 6.1
Specific gravity, maximum	2.20	4.6.2 ASTM D 792
Low temperature flexibility	No cracking	4.6.7.1 -65° ± 2°C (-85° ± 4°F)
Heat resistance, properties after:		4.6.9 350° ± 4°C (662° ± 8°F), 96 hours

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TABLE II. Physical properties. 1/- Continued

Characteristic	Requirement	Test procedure and conditions
Tensile strength, psi (MPa), minimum	2,500 (17.3)	
Ultimate elongation, percent, minimum	200	
Dielectric strength, volts/mil (Kv/mm), minimum	800 (31.5)	
Flammability	Non-flammable	2/
Corrosion	Non-corrosive	2/
Fluid resistance	No change	2/
Fungus resistance	No growth	2/

1/ Unless otherwise specified, all requirements, conditions and procedures are for all classes.

2/ Because of their relative inertness, MIL-I-23053/12 sleevings are not normally subjected to these requirements. However, the acquiring activity (see 6.2.1) may, at its option, specify the performance of any or all of these requirements.

Storage conditions: The supplier shall certify storage at 40°C (104°F) for a period of four years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable polytetrafluoroethylene sleeving is intended for use in areas where resistance to flame and high temperature are required.

Supersedure data: MIL-I-23053/12 Class 5 sleevings have been dropped from this specification, as they are no longer manufactured.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053 /12B	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABL GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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